MSCI World Factor Strategy

Portfolio backtest · 3 October 2014-28 August 2026 · USD

Portfolio rules and benchmark

Table 1. Target allocation

Sleeve	Target	iShares ETF ISIN
Core	60%	IE00B4L5Y983
Momentum	15%	IE00BP3QZ825
Quality	10%	IE00BP3QZ601
Value	15%	IE00BP3QZB59

Weight bands are ±5 percentage points around the targets. A strict breach signals a reset at the next common NAV. The leverage target is 1.25x, with a separate 1.15x-1.35x band that triggers rebalancing at the next common NAV. Actual leverage can move outside the band. The simulation does not model lender margin calls. A weight-only signal preserves debt, a leverage-only signal scales holdings proportionally, and a joint signal resets both.

Committed equity is USD 7.00 million. At 1.25x the initial gross exposure and debt before trading costs are USD 8.75 million and USD 1.75 million. The benchmark is the iShares Core MSCI World UCITS ETF, evaluated with and without modelled borrowing.

Data and cost assumptions

Table 2. Investment cost assumptions

Parameter	Selected value
Financing markup	300 bp over the reference rate
Interest accrual	ACT/360
Platform fee per traded leg	USD 0.85
Investor-side duty	0.15%
One-way spread	5 bp
USD per CHF	1.2368

The common calendar has 3,027 NAV observations and 3,026 return intervals. It uses four iShares USD accumulating NAV histories (BlackRock, 2026a, 2026b, 2026c, 2026d). Financing uses the New York Fed indicative reference series followed by official SOFR (Federal Reserve Bank of New York, 2018, 2026b). Interest is capitalised at NAV dates. A stepped CHF broker commission is included per traded leg.

The model takes the USD 0.85 platform fee and stepped CHF commission schedule from Swissquote Bank Ltd. (n.d., section 1.1). It models investor-side transfer duty at 0.15%, half the 0.30% total rate for foreign securities (Eidgenössische Steuerverwaltung, 2025, paras. 2 and 10). The fixed conversion for CHF commissions is 1.1596 USD/EUR divided by 0.9376 CHF/EUR, using the ECB rates for 31 August 2026 (European Commission, 2026, p. 1).

The model applies a fixed 300 bp markup (see p. 6) and a fixed 5 bp one-way spread per purchase or sale. The baseline includes modelled trading and financing costs and excludes custody fees, external subscriptions, operating costs and additional fund fees. Product expenses are embedded in the NAVs.

Compounded investment performance

CAGR measures growth from committed opening capital over elapsed calendar years, including opening trading costs. The portfolio-minus-Core CAGR gap at the matched 1.25x target is +0.42 percentage points.

Table 3. Unlevered comparison

Investment	CAGR	Volatility	Drawdown	Sharpe	Ending equity, USD m
Factor portfolio	11.87%	14.68%	-33.56%	0.691	26.59
MSCI World Core	11.52%	14.90%	-34.07%	0.662	25.62

Table 4. Matched 1.25x leverage target

Investment	CAGR	Volatility	Drawdown	Sharpe	Ending equity, USD m
Factor portfolio	13.24%	18.25%	-39.70%	0.655	30.76
MSCI World Core	12.82%	18.59%	-40.36%	0.627	29.43

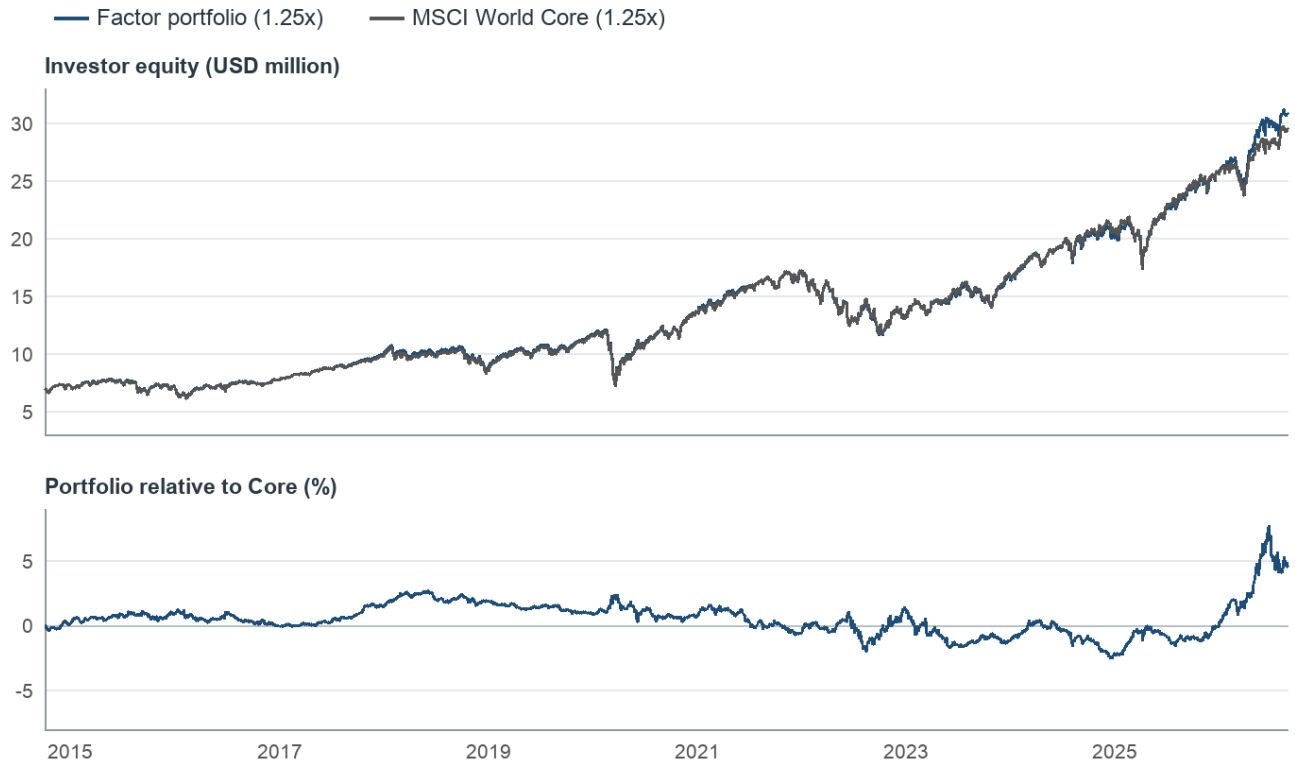


Figure 1. Investor equity at the matched 1.25x target. Relative wealth is 100 × (portfolio equity / Core equity - 1).

Drawdown and relative wealth

Drawdown uses the running equity high-water mark, with committed opening capital as the initial floor. Volatility uses 252-observation annualisation. The Sharpe ratio uses excess returns over the reference rate.

Table 5. Factor portfolio at 1.25x versus unlevered Core (1x)

Investment	Target	CAGR	Volatility	Sharpe
Factor portfolio	1.25x	13.24%	18.25%	0.655
MSCI World Core	1x	11.52%	14.90%	0.662

Drawdowns at a matched 1.25x target



Figure 2. Drawdowns from committed-capital high-water marks. Both investments target 1.25x.

Table 6. Relative wealth at selected endpoints

Last common NAV	Portfolio relative to Core, 1.25x
2022-12-30	+1.21%
2023-12-29	-1.17%
2024-12-31	-2.19%
2025-12-31	-0.07%
2026-08-28	+4.51%

The checkpoints are read from the continuing full-period accounts.

Quality and Core return correlation

The Pearson correlation between iShares Core and Quality returns is 0.985513 (0.986 rounded). The calculation uses the 3,027 common iShares NAV dates from 3 October 2014 to 28 August 2026 (BlackRock, 2026a, 2026c). For each ETF, the return is the current USD NAV divided by the previous common-date NAV minus one. The first date provides opening NAVs, leaving 3,026 paired returns for the correlation.

Paired specifications and bootstrap intervals

The 14 cases include the baseline and one-at-a-time changes to start date, financing markup, fund fee, leverage band and sleeve band. The five later starts advance one to five iShares NAV observations. Earlier starts fall outside the sample. Both accounts target 1.25x. The baseline charges no additional fund fee. The two fee variants charge both accounts 2.5 or 5 bp per year.

Table 7. Portfolio minus Core CAGR across paired specifications

Endpoint	Baseline pp	Median pp	Minimum pp	Maximum pp	Positive
2023-12-29	-0.14	-0.14	-0.34	-0.06	0/14
2024-12-31	-0.24	-0.24	-0.47	-0.18	0/14
2025-12-31	-0.01	-0.01	-0.29	+0.05	5/14
2026-08-28	+0.42	+0.42	+0.17	+0.47	14/14

The baseline gap was -0.24 percentage points at the 2024 year-end and +0.42 at the August 2026 endpoint. Positive gaps rose from 0 of 14 cases at the 2023 and 2024 year-ends to 5 of 14 at the 2025 year-end and 14 of 14 in August 2026.

Table 8. Conditional stationary bootstrap of the 1.25x CAGR gap

Mean block	Draws	Gap pp	95% low pp	95% high pp
20	5,000	+0.42	-0.72	+1.59
60	5,000	+0.42	-0.69	+1.71

The stationary bootstrap (Politis & Romano, 1994, pp. 1303-1304) resamples paired realised account returns, with opening costs counted once per draw. For both mean block lengths, the central 95% of resampled CAGR gaps spans negative and positive values and includes zero. The resampled returns already reflect the trading rules applied to the historical paths.

Weight-band comparison

The capped-relative rule uses the smaller of 5 percentage points and 20% of each sleeve target.

Table 9. Absolute and capped-relative weight bands

Rule	Target	CAGR	Volatility	Drawdown
Absolute / separate leverage	1x	11.87%	14.68%	-33.56%
Capped-relative comparison	1x	11.79%	14.71%	-33.72%
Absolute / separate leverage	1.25x	13.24%	18.25%	-39.70%
Capped-relative comparison	1.25x	12.92%	18.13%	-39.89%

Both band rules were evaluated retrospectively. The wider bands delivered higher CAGR in this sample, while the narrower bands had slightly lower volatility at the 1.25x target.

Product comparisons

Dimensional at the matched 1.25x leverage target

The comparison uses the daily USD NAV of the Dimensional Global Core Equity Fund USD Accumulation share class (IE00B2PC0153), supplied through Finanzfluss (Morningstar, n.d.). The common iShares and Dimensional sample contains 2,929 NAV dates from 3 October 2014 to 28 August 2026.

Table 10. Portfolio and Dimensional at a 1.25x target

Investment	CAGR	Volatility	Drawdown
Factor portfolio	13.24%	18.59%	-39.70%
Dimensional	11.32%	20.92%	-43.55%

Both accounts use a 1.25x target with the same leverage adjustment band and modelled trading and financing costs. The Dimensional account holds one fund. The main portfolio and Core comparison uses its own iShares-only calendar.

Amundi 2x product comparison

The comparison uses 232 dates common to the four iShares NAVs and the official USD NAV of Amundi MSCI World (2x) Leveraged UCITS ETF Acc (FR0014010HV4), from 30 September 2025 to 31 August 2026 (Amundi Asset Management, 2026a). The NAV data are in the supplied workbook, sheet “Nav History”, column “Official NAV”.

Table 11. Portfolio and Amundi over the common NAV window

Investment	Cumulative return	Volatility	Drawdown
Factor portfolio, 2x target	36.38%	26.22%	-18.74%
Amundi 2x ETF NAV	27.51%	23.79%	-17.50%

The portfolio is simulated on the full iShares calendar within the comparison window. Its equity is sampled on common dates. The portfolio uses band-managed leverage. The Amundi series uses the ETF’s reported NAV. Both paths are normalised to the first common post-entry observation. This normalisation excludes the modelled opening charge from the displayed cumulative return.

The Amundi ETF tracks an index with daily 2x leverage, and its NAV reflects borrowing costs and product expenses (Amundi Asset Management, 2026b, pp. 1-3). Investor trading charges on Amundi are outside the series.

Financing markup scenarios

The project brief records a 6.65% USD borrowing rate from a private Swissquote account snapshot dated 26 September 2026. The New York Fed 30-day SOFR average was 3.705% on 25 September (Federal Reserve Bank of New York, 2026a, 25 September observation). The 294.5 bp difference was rounded to a fixed 300 bp markup. The account category and historical broker rates remain unverified.

Table 12. Portfolio CAGR by financing markup at a 1.25x target

Markup above reference, bp	Full-period CAGR	Later-period CAGR
0	13.64%	13.30%
300	13.24%	12.61%
500	12.78%	12.13%
1000	11.41%	11.11%

The full period starts on 3 October 2014. The later period restarts with committed opening equity on 4 January 2022. Both end on 28 August 2026. Each row is a separate simulation. The reference rate applies in every scenario, including at a zero markup.

Sources and reproducibility

The market data, cost and method references are on the final page. The iShares and New York Fed exports were captured on 27 and 28 September 2026, respectively. The Amundi workbook identifies its update as 25 September 2026. The ZIP contains the workbook used in the comparison under Inputs/market_data/amundi. The original workbook download URL and retrieval date are unverified.

Interest accrues for calendar days from the prior NAV date up to but excluding the current NAV date, using the latest available reference rate. The ACT/360 day count and rate carry-forward follow the conventions described for the SOFR Averages and Index (Federal Reserve Bank of New York, n.d., SOFR Averages and Index section).

The accompanying calculation files record source checksums, the computed results, the cases and the plotted series.

Appendix. Supplementary calculations for the video

1. Unlevered Quality allocation test

The test uses the 3,027 common iShares NAVs from 3 October 2014 to 28 August 2026, with the same USD 7 million starting equity and trading-cost assumptions as the main backtest. There is no borrowing. In Core / Momentum / Quality / Value order, the target weights change from 60% / 15% / 10% / 15% to 60% / 10% / 20% / 10%.

Within each comparison both allocations follow the same rule. The earlier rule caps each band at the smaller of 5 percentage points and 20% of the sleeve target. The final rule uses absolute 5-percentage-point bands. A strict breach signals a reset at the next common valuation.

Table 13. Quality allocation under each rebalancing rule

Rule	CAGR at 10%	CAGR at 20%	CAGR change, pp	Max. drawdown at 10%	Max. drawdown at 20%
Capped-relative	11.79%	11.68%	-0.11	-33.72%	-33.65%
Absolute	11.87%	11.63%	-0.24	-33.56%	-33.54%

The roughly 0.1-percentage-point reduction described in the video matches the earlier capped-relative comparison. Using the unrounded values, maximum drawdown is 0.06465 percentage points shallower under that rule. The comparison includes the corresponding changes to Momentum and Value.

2. Hypothetical fund assets after ten years

The steady-acquisition model starts with USD 7 million in founder capital, assumes a fixed 6% annual return before the fund fee and uses USD 250,000 per new client equivalent. Scheduled new client equivalents rise from 2 in year 1 to 20 in year 10 in steps of 2. Gross subscriptions total USD 27.5 million. Year-one subscriptions start in month 7. The model assumes 10% annual attrition of external units.

The 0.30% annual fund fee applies to this projection. The model converts return and attrition to monthly rates. Each month it applies investment return and charges one twelfth of the annual fund fee on opening assets, then processes redemptions and subscriptions. Year-ten fund assets are the sum of the closing founder and external holdings.

Table 14. Closing fund assets after 120 months, USD million

Annual fund fee	Founder equity	External equity	Total fund assets
0.30%	12.17	23.30	35.47

Under these assumptions, we expect about USD 36 million in fund assets after ten years.

Reproduction files: Evidence/results.json, section video_supplement, records full-precision results. Code/config/steady_acquisition_video.json records the model inputs. The package runner rebuilds both calculations.

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